and become ever more spread out, diluted, and barren. But quintessence provides accelerated expansion that at some point draws to a close, suggesting a far future less bleak and desolate than that following from accelerated expansion that's eternal. By measuring changes in the acceleration of space over long time spans (through observations of supernovae at various distances and hence at various times in the past), SNAP may be able to distinguish between the two possibilities. By determining whether the dark energy truly is a cosmological constant, SNAP will give insight into the long-term fate of the universe.

Space, Time, and Speculation

The journey to discover the nature of space and time has been long and filled with many surprises; no doubt it is still in its early stages. During the last few centuries, we've seen one breakthrough after another radically reshape our conceptions of space and time and reshape them again. The theoretical and experimental proposals we've covered in this book represent our generation's sculpting of these ideas, and will likely be a major part of our scientific legacy. In Chapter 16, we will discuss some of the most recent and speculative advances in an effort to cast light on what might be the next few steps of the journey. But first, in Chapter 15, we will speculate in a different direction.

While there is no set pattern to scientific discovery, history shows that deep understanding is often the first step toward technological control. Understanding of the electromagnetic force in the 1800s ultimately led to the telegraph, radio, and television. With that knowledge, in conjunction with subsequent understanding of quantum mechanics, we were able to develop computers, lasers, and electronic gadgets too numerous to mention. Understanding of the nuclear forces led to dangerous mastery over the most powerful weapons the world has ever known, and to the development of technologies that might one day meet all the world's energy needs with nothing but vats of salt water. Could our ever deepening understanding of space and time be the first step in a similar pattern of discovery and technological development? Will we one day be masters of space and time and do things that for now are only part of science fiction?